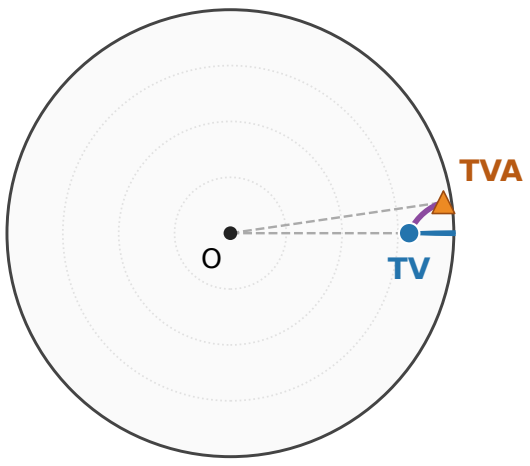
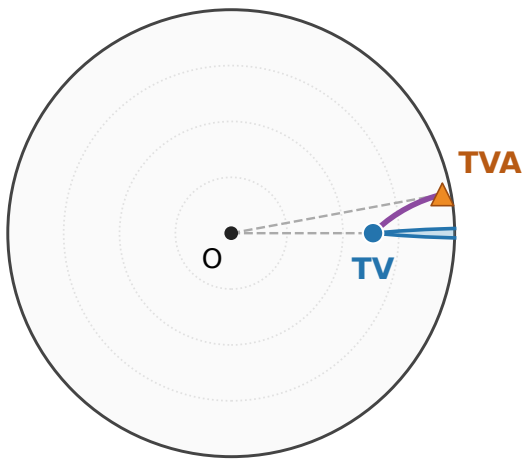
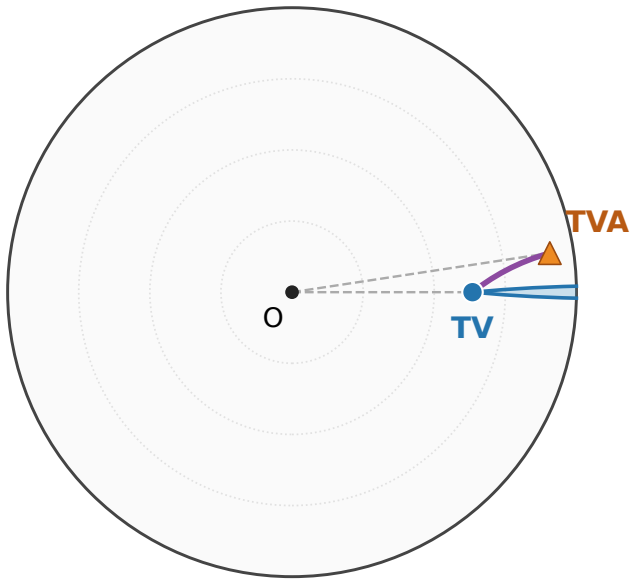
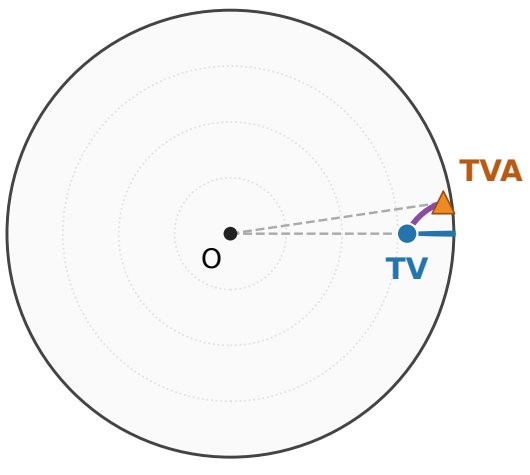
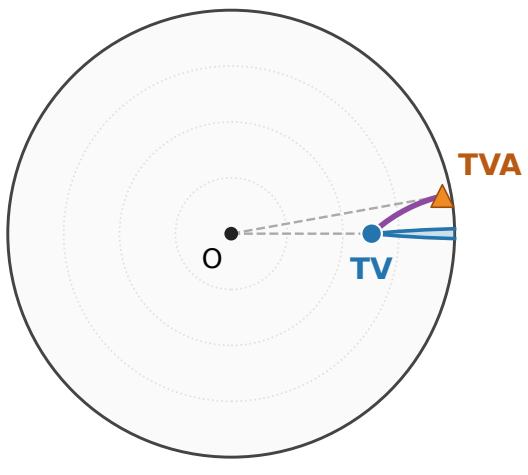
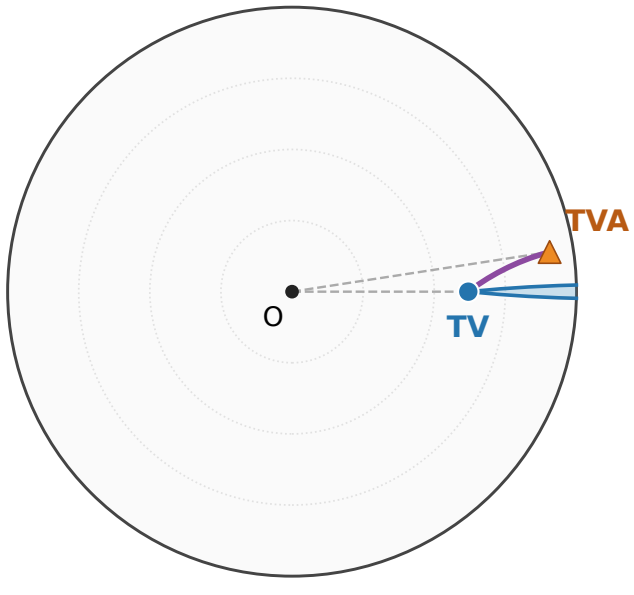
Actual Poincare-ball examples: TV -> TVA

Same three validation videos across models | real 512D features | epoch 30 | seed 50

Each panel is the exact O-parent-child hyperbolic plane, rotated so the parent is on the positive horizontal axis.

	Original C0 lambda=.1, c=1, tau=.07 Ball radius R=1.000	Current fixed lambda=.2, c=1, tau=.035 Ball radius R=1.000	Current learned lambda=.2, c=.617123, tau=.043651 Ball radius R=1.273
Sample 1 validation #1463	 r(TV)=2.197 r(TVA)=3.984 d(parent, child)=2.141 psi=2.58 deg phi=67.14 deg OUTSIDE cone; excess=64.56 deg	 r(TV)=1.498 r(TVA)=3.911 d(parent, child)=2.559 psi=5.40 deg phi=44.65 deg OUTSIDE cone; excess=39.25 deg	 r(TV)=1.906 r(TVA)=3.990 d(parent, child)=2.217 psi=5.40 deg phi=38.42 deg OUTSIDE cone; excess=33.02 deg
Sample 2 validation #1598	 r(TV)=2.149 r(TVA)=3.979 d(parent, child)=2.161 psi=2.71 deg phi=65.06 deg OUTSIDE cone; excess=62.35 deg	 r(TV)=1.477 r(TVA)=3.886 d(parent, child)=2.543 psi=5.53 deg phi=42.72 deg OUTSIDE cone; excess=37.19 deg	 r(TV)=1.844 r(TVA)=3.985 d(parent, child)=2.266 psi=5.71 deg phi=37.16 deg OUTSIDE cone; excess=31.45 deg
Sample 3 validation #1611	 r(TV)=2.221 r(TVA)=3.979 d(parent, child)=2.013 psi=2.52 deg phi=57.47 deg OUTSIDE cone; excess=54.96 deg	 r(TV)=1.522 r(TVA)=3.852 d(parent, child)=2.446 psi=5.26 deg phi=39.89 deg OUTSIDE cone; excess=34.64 deg	 r(TV)=1.931 r(TVA)=3.984 d(parent, child)=2.161 psi=5.29 deg phi=34.60 deg OUTSIDE cone; excess=29.31 deg

Parent embedding Child embedding Parent-child geodesic Parent entailment cone

Coordinates are curvature-aware Poincare coordinates; the joint-learning model has a larger ball because its c is smaller.

The parent-child geometry is preserved in each slice. Absolute orientations across different panels do not describe a global 2D embedding.